Cervical and Lumbar Disk Replacement in Athletes: Is It Safe to Return to Play? A Systematic Review of the Scientific Literature and Lay Press

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BACKGROUND AND OBJECTIVES: Cervical/lumbar total disk replacements (TDRs) are often performed for degenerative conditions but rarely in athletes. Therefore, we sought to conduct a systematic review of athletes undergoing TDRs of both the scientific literature and lay press, with an emphasis on contact sport athletes.

METHODS: In accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analysis guidelines, PubMed/Embase/Cochrane/Scopus/Web of Science databases were queried for all primary literature reporting TDRs in athletes, including both cervical/lumbar disk replacements (CDRs/LDRs). Sports were dichotomized into contact vs limited/noncontact. Because of the public nature of injuries in elite athletes, the lay press was also queried.

RESULTS: A total of 488 scientific studies were screened, of which 10 met inclusion criteria. Cervical: seven studies reported CDRs in 53 athletes, of which 7 were professional, 22 semiprofessional, and 24 recreational. Of the seven professional athletes, there was one contact sport athlete (kickboxer). All 7/7 professional and 21/22 semiprofessional athletes successfully returned-to-play at 8-52 weeks without complication. Lumbar: Three studies discussed LDRs in 51 athletes, of which 17 were professional, 6 semiprofessional, and 28 recreational. Of the 17 professional athletes, eight played contact sports (2 boxing, 2 alpine skiing, 2 soccer, judo, rugby). All 17 professional and 6 semiprofessional athletes successfully returned to play at 9–21 weeks. Lay Press: five professional contact sport athletes underwent CDRs, and all returned to play: 3 hockey, 1 mixed-martial arts, and 1 Australian-rules football.

CONCLUSION: The scientific literature and lay press revealed 14 professional contact sport athletes who underwent TDR—6 CDRs and 8 LDRs—all with successful return to play. From the little data that exist, it seems that TDR may be safe in elite athletes; however, the small number of patients highlights the major paucity of data on the safety of TDR in elite contact sport athletes.

KEY WORDS: Total disk replacement, Arthroplasty, Return to play, Athlete, Spine surgery

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Total disk replacements (TDRs), either a cervical disk replacement (CDR) or lumbar disk replacement (LDR), are performed for degenerative conditions of the spine. Total disk replacements have shown equivalence in safety and effectiveness to fusions in both the cervical[1-4] and lumbar regions.[5-10] Select studies have demonstrated superior outcomes to their fusion alternatives, with better disease-specific outcomes,[3] lower rates of adjacent segment disease,[2] and higher preservation of mobility.[8] However, studies, to date, are limited to the general population, and literature on return to play (RTP) after TDR in athletes remains sparse.[11,12]

Among the few studies that have investigated outcomes after TDRs in athletes, the overall level of evidence is low.[11-15] Current studies include mostly case reports[13] and reviews,[11] with some studies focusing exclusively on TDRs,[15] whereas others have included a mixed cohort of TDRs and fusions.[14] The heterogeneous nature of these studies calls for an effort to systematically organize RTP data in athletes undergoing TDRs. Moreover, CDR

ABBREVIATIONS: ACDF, anterior cervical diskectomy and fusion; **CDR,** cervical disk replacement; **LDR,** lumbar disk replacement; **LP,** Lumbar Prosthesis; **LP–ESP,** Lumbar Prosthesis–Elastic Spine Pad; **MMA,** mixed martial arts; **NFL,** National Footbal League; **NHL,** National Hockey League; **RTP,** return to play; **TDR,** total disk replacement.

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in elite athletes has become a controversial topic in recent years, garnering the most attention in professional hockey.[16]

Given the lack of scientific data yet growing public interest surrounding TDR in athletes, we aimed to study this topic further. The current systematic review sought to (1) summarize scientific reports of athletes undergoing TDRs, (2) report RTP data of athletes after TDR, and (3) query the lay press for reports of elite athletes undergoing TDR.

METHODS

Literature Search Strategy

We conducted a systematic review to evaluate RTP in athletes after cervical/lumbar TDR. Search strings for PubMed/Embase/Cochrane/Scopus/Web of Science databases were devised for studies published between 9/1/1970 and 9/30/2022. Full search terms are presented in **Supplementary Table 1**, http://links.lww.com/NEU/D904.

The lay press was queried using a combination of international internet searches across several different sports. Keywords and search terms from **Supplementary Table 1**, http://links.lww.com/NEU/D904, were used. Although not a conventional part of a scientific systematic review, our a priori study objectives would not be met if we did not include public and widely known cases of TDRs in elite athletes.

Selection Criteria

Studies were included if they discussed TDR in athletes in any capacity and had their full text available in English. Study exclusion criteria included nonprimary literature, studies without TDR, and studies of nonathletes. Studies were deduplicated and selected using Covidence. The abstract and full-text screening were performed independently by two reviewers (authors J.J. and A.D.D.) in Covidence. Disagreements were resolved by a third author (A.M.H.). Publications were included for full-text screening if the title or abstract included any information about TDR in athletes.

Given the sparse literature surrounding the topic of TDR in athletes, the inclusion criteria were kept broad and were (1) athletes with different levels of participation and contact level and (2) studies with cohorts containing athletes who received various spine surgeries as long as they included any athlete who underwent a TDR in their sample. In addition, we looked at several large studies (N = 300-5000)[2,4,5,7,8] that investigated outcomes after TDR that did not specify "athletes" in their title and/or abstracts to ensure that no subpopulations were missed. None of these general populations made mention of semiprofessional or professional athletes.

Data Extraction

The following variables were extracted from each study: basic study information and demographics, diagnosis and recovery (diagnosis, symptoms, approach, levels, segment, prosthesis), and outcomes regarding RTP (follow-up time, symptom resolution status, time to RTP to practice/competition, radiographic findings at follow-up, complications, reoperation).

Data Analysis

Descriptive statistics were performed. Contact level was defined by previously published criteria by Rice et al[17] and was categorized into (1) contact (eg, boxing, basketball, football), (2) limited-contact (eg, baseball, skating, volleyball), and (3) noncontact sports (eg, running, tennis, powerlifting). The level of evidence/risk of bias was assessed, and quality appraisal was performed using the Level of Evidence for Primary Research Question.[18] The current study followed the Preferred Reporting Items for Systematic Reviews and Meta-Analysis guidelines.[19] The study was not registered on a systematic reviews database (eg, The International Prospective Register of Systematic Reviews).

Search of the Lay Press for Professional Athletes Undergoing TDR

Given the plethora of publicly available information on professional athletes who have undergone TDR, we performed an in-depth, international, online search to summarize all professional athletes who have undergone a TDR. Internet, news, and social media outlets were queried. North American and international news outlets were searched. Because TDRs were less controversial in noncontact sport athletes compared with contact sport athletes, the search focused on contact sport athletes. An exhaustive list of contact sports included the following: American football, Australian-Rules football, basketball, boxing, cheerleading, diving, field hockey, gymnastics, ice hockey, lacrosse, martial arts, rodeo, rugby, downhill skiing, ski jumping, snowboarding, soccer, handball, ultimate frisbee, water polo, and wrestling.[17] The final list of athletes from the search was compared with that of athletes from the scientific literature to ensure no overlap.

RESULTS

Descriptive Statistics

A total of 488 studies were screened, and 10 met inclusion criteria.[13-15,20-24] Seven studies[13-15,20,21,25,26] discussed athletes who had undergone CDR, and three studies[22-24] discussed those with LDR. A full Preferred Reporting Items for Systematic Reviews and Meta-Analysis flow diagram is presented in Figure 1.

I. Cervical Disk Replacement

Study Information and Demographics

Of the seven articles, three studies[14,20,21] were retrospective cohorts, three were case reports,[13,25,26] and one was a prospective cohort.[15] Four studies[13,15,25,26] investigated exclusively those with CDR, whereas the remaining studies included patients with CDR, fusions, and decompressions only. A total of 53 athletes undergoing CDR were discussed across the seven studies, of which seven (13%) were professional, 22 (42%) semiprofessional, and 24 (45%) recreational (Figure 2A). Of the seven professional athletes, one (14%) played contact sports (kickboxing), three (43%) played limited-contact sports (3 baseball), and three (43%) played noncontact sports (2 lugers, 1 marathoner) (Figure 2B). Two studies[25,26] included a semiprofessional athlete each, both of whom were collegiate football players. One study included athletes of both semiprofessional and recreational levels, and sports/contact level was not specified.[15] Across all patients, the mean age ranged from 21.0 to 40.0 years (Table 1).[13-15,20,21]

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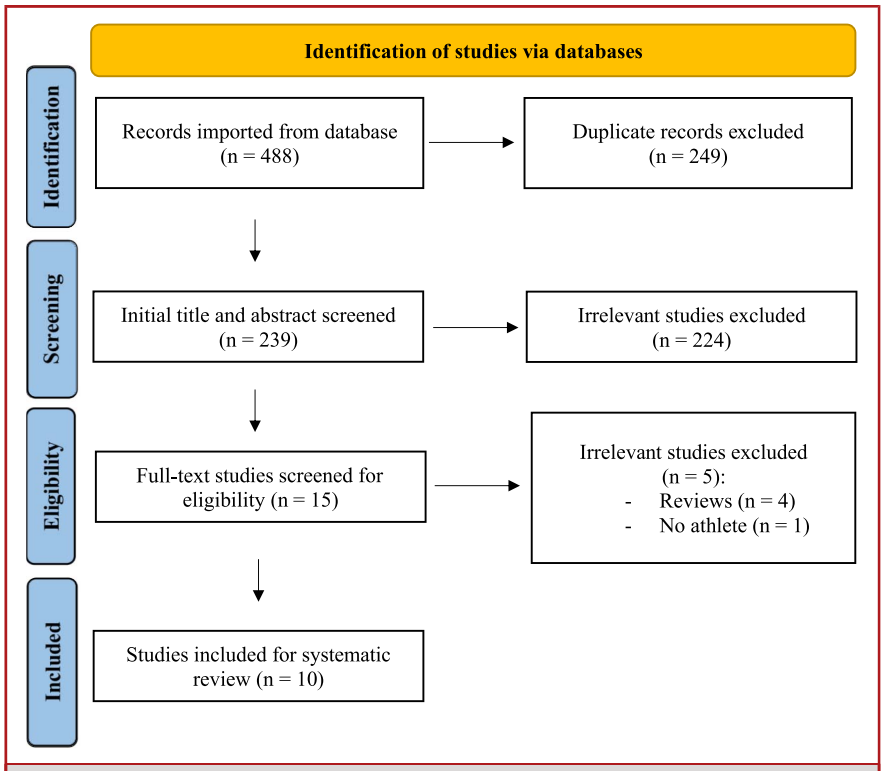

FIGURE 1. *Preferred Reporting Items for Systematic Reviews and Meta-Analyses flow diagram.*

Diagnosis and Surgery

Across seven studies, all patients were diagnosed with disk herniation/degeneration with symptoms of radicular/axial pain. One level was most commonly reported, besides one study[25] that reported a 2-level CDR. Only one study[15] specified the prosthesis used (Prestige Lumbar Prosthesis) (Table 2).[13-15,20,21]

Outcomes and RTP

Across seven studies, the mean follow-up ranged widely from 0.1 to 7.0 years. Three studies[13,25] reported their athlete's symptom resolution as "complete" and another[15] commented that 31/45 patients reported that they were "very satisfied" with their outcome. All seven professional and 21/22 semiprofessional athletes successfully returned to practice after a mean of 3–6 weeks and returned to their previous level of competition after a mean of 8–52 weeks. The single professional contact sport athlete was a kickboxer who underwent a CDR in the final years of her career, postoperatively played 1 year, and retired as planned.[13] No professional American football player underwent a CDR. One American collegiate football player successfully returned to play after a 1-level CDR at 8 weeks without complication. Another

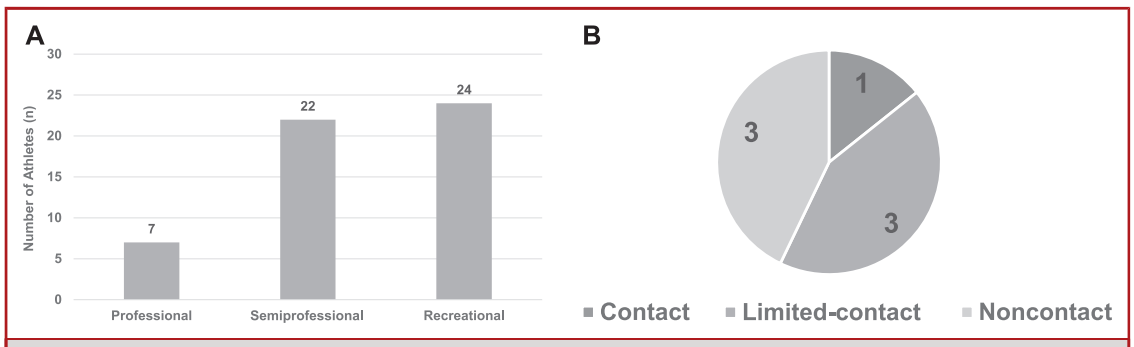

FIGURE 2. **A**, *Level of sport/activity of athletes with CDR.* **B**, *Level of contact of professional athletes with CDR. CDR, cervical disk replacement.*

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TABLE 1. CDR in Athletes: Demographics

Author, y	Country	Study type	N	N (TDR)	Sport/activity level	Contact	Age (mean)	M/F
Lindi-Sugino et al,[13] 2021	Brazil	Case report	1	1	Professional (kickboxing)	Contact	34.0	F
Mai et al,[14] 2018	US	Retrospective cohort	101	2	Professional (2 baseball)	Limited-contact	—	M
Roberts et al,[20] 2011	US	Retrospective cohort	40	1	Professional (baseball)	Limited-contact	30.7[a]	M
Shah et al,[21] 2019	India	Retrospective cohort	10	1	Professional (marathon)	Noncontact	32.4[a]	—
Reinke et al,[15] 2017	Germany	Prospective cohort	50	50	Professional (2 lugers) Semiprofessional (20 unspecified) Recreational (24 unspecified) Low activity (4 unspecified)	Professional: noncontact (2)	40.0	28/22
Satalich et al,[26] 2022	United States	Case report	1	1	Semiprofessional (football)	Contact	23.0	M
Satalich et al,[25] 2022	United States	Case report	1	1	Semiprofessional (football)	Contact	21.0	M

CDR, cervical disk replacement; TDR, total disk replacement.
[a]Average represents that of patients with and without TDR (eg, ACDF, microdiskectomy).
—, Not reported/insufficient information in the study.

American collegiate football player underwent a 2-level CDR, yet no RTP information was available because the study was performed only at 3-weeks postoperatively. Of the 24 recreational athletes, 20 (83%) returned to practice at mean 4 weeks and returned to play at mean 25 weeks.[15] Four (17%) recreational athletes did not RTP, for reasons not specified.[15] One athlete showed anterior calcification at the level of surgery at last follow-up.[13] No future injuries, complications, or reoperations were reported during the follow-up period (Table 3).[13-15,20,21,25,26]

II. Lumbar Disk Replacement

Study Information and Demographics

All three articles originated from outside the United States Russia,[22] Australia,[23] Germany.[24] They included a case report,[23] a retrospective cohort of 11 patients,[22] and a prospective cohort of 39 patients.[24] All three studies investigated exclusively those with lumbar TDR.[22-24] A total of 51 athletes with lumbar TDR were studied, of which 17 (33%) were professional, 6 (12%) semiprofessional, and 28 (55%) recreational (Figure 3A). Of the 17 professional athletes, eight

TABLE 2. CDR in Athletes: Diagnosis and Surgery

Author, y	Diagnosis	Symptoms	Approach	Levels	Segment	Prosthesis
Lindi-Sugino et al,[13] 2021	Disk herniation	Radicular pain	Lateral	1	C5–C6	—
Mai et al,[14] 2018	Disk herniation	—	—	1	—	—
Roberts et al,[20] 2011	Disk herniation	—	—	1	—	—
Shah et al,[21] 2019	Disk herniation	Radicular/axial pain	Anterior	1	C5–C6	—
Reinke et al,[15] 2017	Disk herniation	Radicular pain	Anterior	1	—	Prestige LP
Satalich et al,[26] 2022	Disk herniation	Radicular pain	—	1	C6–C7	—
Satalich et al,[25] 2022	Disk herniation	Radicular pain	Anterior	2	C5–C6 C6–C7	—

CDR, cervical disk replacement; LP, Lumbar Prosthesis.
—, Not reported/insufficient information in the study.

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TABLE 3. CDR in Athletes: Outcomes and RTP

Author, y	Follow-up (mean, y)	Outcomes/symptom resolution	RTP to practice (mean, wk)	RTP to Competition (mean, wk)	Radiographic	Complication	Reoperation
Lindi-Sugino et al,[13] 2021	7.0	Complete	3	12	Anterior calcification at level	None	None
Mai et al,[14] 2018	6.1	—	—	36	—	None	None
Roberts et al,[20] 2011	5.1[a]	—	—	52[a]	—	None	None
Shah et al,[21] 2019	3.8	—	6	12	—	None	None
Reinke et al,[15] 2017	4.4	Very satisfied (31/45) Satisfied (11/45) Not satisfied (3/45)	4	25	No abnormalities	None	None
Satalich et al,[26] 2022	0.8	Complete	4	8	No abnormalities	None	None
Satalich et al,[25] 2022	0.1	Complete	—	—	No abnormalities	None	None

CDR, cervical disk replacement; RTP, return to play.
[a]Average represents that of patients with and without TDR (eg, ACDF, microdiskectomy).
—, Not reported/insufficient information in the study.

(47%) played contact sports (2 boxing, 2 alpine skiing, 2 soccer, 1 judo, 1 rugby), four (24%) played limited-contact sports (2 weightlifting, 1 figure skating, 1 skiing), and five (29%) played noncontact sports (2 fitness, 1 ballet, 1 archery, 1 tennis) (Figure 3B). Of the six semiprofessionals, one (17%) played limited-contact (1 skiing) and five (83%) noncontact sports (3 running, 1 tennis, 1 golf).[24] Only one study included athletes of recreational level, and sports/contact level was not specified.[24] Across all patients, the mean age ranged from 30.0 to 39.8 years, and 57% were male (Table 4).[22-24]

Diagnosis and Surgery

Across three studies, 41 (80%) patients were diagnosed with degenerative disk disease, nine (18%) with previous diskectomy, and one (2.0%) with adjacent-level disease after previous fusion. All had radicular/back pain. Surgery involved 32 anterior retroperitoneal, 7 transperitoneal, and 1 posterior approach. Of 51 patients, 48 were one-level and 3 were 2-level LDRs. All surgeries involved level(s) L4–L5 and/or L5–S1. All three studies specified the prosthesis used (M6-L, Lumbar Prosthesis–Elastic Spine Pad, ProDisc II) (Table 5).[22-24]

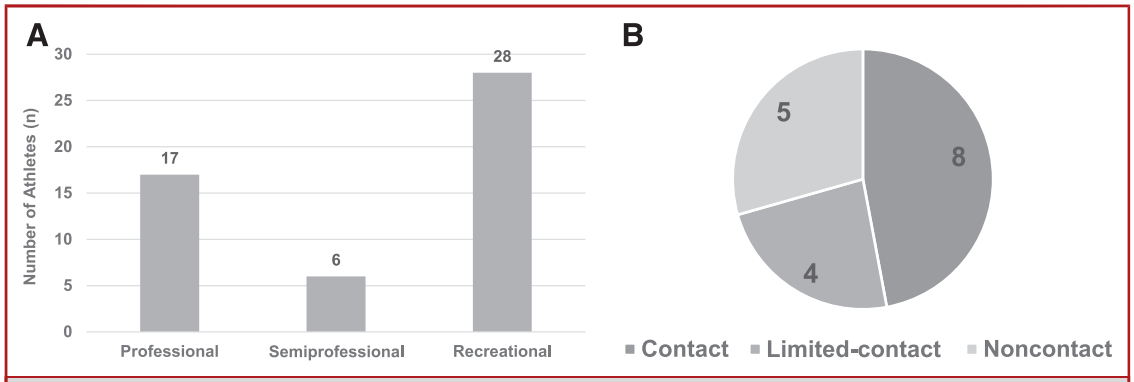

FIGURE 3. **A**, *Level of sport/activity of athletes with LDR.* **B**, *Level of contact of professional athletes with LDR. LDR, lumbar disk replacement.*

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Table 4. LDR in athletes: demographics

Authors and year	Country	Study type	N	N (TDR)	Sport/activity level	Contact	Age (mean)	M/F
Byvaltsev et al,[22] 2021	Russia	Retrospective cohort	11	11	Professional (2 alpine skiing, 2 boxing, 2 weightlifting, judo, figure skating, soccer, tennis, archery)	Contact (6) Limited-contact (3) Noncontact (2)	31.2	7/4
Gan et al,[23] 2020	Australia	Case report	1	1	Professional (rugby)	Contact	30.0	M
Siepe et al,[24] 2007	Germany	Prospective cohort	39	39	Professional (soccer, ballet, 2 fitness, skiing) Semiprofessional (3 running, skiing, tennis, golf) Recreational (28 unspecified)	Professional: Contact (1) Limited-contact (1) Noncontact (3) Semi-professional: Limited-contact (1) Noncontact (5)	39.8	21/18

LDR, lumbar disk replacement; TDR, total disk replacement.
—, not reported/insufficient information in the study.

Outcomes and RTP

Across three studies, the mean follow-up ranged from 2.0 to 3.2 years. One study[24] commented that 33/39 patients reported that they were "very satisfied" with their outcome. All 17 professional athletes and 6 semiprofessional athletes successfully returned to practice after a mean of 12–14 weeks and returned to play to their previous level of competition after a mean of 9.7–20.8 weeks. No separate information was provided on recreational athletes.[24] A total of 5/39 (12.8%) patients had complications, including heterotopic ossification, adjacent-level disease, two cases of retrograde ejaculation, and two cases of L5 radiculopathy. One study[24] noted 13 cases of subsidence at the last follow-up and reported two reoperations, including one revision surgery for abdominal hematoma and one diskectomy for adjacent segment herniation (Table 6).[22-24]

Level of Evidence/Quality Appraisal

All 10 studies were therapeutic studies investigating outcomes after cervical/lumbar TDR. Of those, three studies had level III evidence and seven had level IV evidence. All level III studies were retrospective cohorts with comparative group(s), and level IV studies included case reports, retrospective cohorts, and prospective cohorts that did not have any comparison group (**Supplementary Table 2**, http://links.lww.com/NEU/D905).

III. Search of the Lay Press

All athletes found in the lay press search had undergone a CDR. The final search yielded six professional athletes: three hockey players,[16,27,28] one mixed-martial arts fighter,[29] one cricket player,[30] and one Australian-rules football player.[31] Of the six professional athletes, five were contact sport athletes and one was limited contact (cricket). All athletes had a successful RTP at their previous level of competition. Noncontact athletes included a golfer[32] and a swimmer,[33] all of whom had undergone CDRs with documented RTP.

Among five professional contact sport athletes, three were hockey players in the National Hockey League (NHL). Jack

TABLE 5. LDR in Athletes: Diagnosis and Surgery

Authors and year	Diagnosis	Symptoms	Approach	Levels	Segment	Prosthesis
Byvaltsev et al,[22] 2021	Degenerative disk	Radicular pain	Posterior	1	L4–L5 (5) L5–S1 (6)	M6-L
Gan et al,[23] 2020	Degenerative disk	Radicular/back pain	—	1	L4–L5	LP-ESP
Siepe et al,[24] 2007	Degenerative disk (29) Previous diskectomy (9) Adjacent level disease s/p previous fusion (1)	Back pain	Anterior retroperitoneal (32) Transperitoneal (7)	1 (36) 2 (3)	L4–L5 (10) L5–S1 (26) L4–L5/L5–S1 (3)	ProDisc II

LDR, lumbar disk replacement; LP–ESP, Lumbar Prosthesis–Elastic Spine Pad.
—, Not reported/insufficient information in the study.

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TABLE 6. LDR in Athletes: Outcomes and RTP

Authors and year	Follow-up (mean, y)	Outcomes/symptom resolution	RTP to practice (mean, wk)	RTP to competition (mean, wk)	Radiographic	Complication	Reoperation
Byvaltsev et al,[22] 2021	3.2	—	—	9.7	—	Heterotopic ossification (1) Retrograde ejaculation (1)	None
Gan et al,[23] 2020	2.0	—	12	16	—	None	None
Siepe et al,[24] 2007	2.2	Very satisfied (33) Satisfied (4) Not satisfied (2)	12–24	20.8	Subsidence (13)	Adjacent-level disease (1) Retrograde ejaculation (1) L5 radiculopathy (2)	Revision surgery for hematoma of abdomen (1) Diskectomy for adjacent segment herniation (1)

LDR, lumbar disk replacement; RTP, return to play.
—, Not reported/insufficient information in the study.

Eichel of Buffalo Sabres was the first among the NHL athletes to undergo a CDR for disk herniation.[16,27,29] Although there was an initial drop in the total number of games played during the season immediately after his surgery on 11/04/21, Jack continued to demonstrate successful RTP, with 67 games played in latest season with comparable stats from before the surgery.[34] Tyler Johnson underwent a CDR on 12/03/21 for neck pain reportedly dating back to an injury 3 months earlier after a head-to-body contact during play.[35] He, similar to Jack, initially had a reduction in the total number of games played during recovery; however, his performance seems to have stabilized, with 56 total games played in the latest season.[34] Finally, Joel Farabee[28] underwent a CDR for disk herniation on 06/24/22 and since has demonstrated successful RTP, playing 52 games during the latest season.[34] Detailed performance statistics of these athletes are presented in **Supplemental Table 3a-c**, http://links.lww.com/NEU/D906.

The remaining two athletes were Chris Weidman, an American professional mixed martial artist and a former middleweight champion of the Ultimate Fighting Championships (UFC),[36] and Nathan Jones, a former Australian-rules football player who played for the Melbourne Football Club in the Australian Football League.[31,37] Chris Weidman received a CDR in 06/2016 for disk herniation, before returning to his first fight back to the UFC in the same year (11/12/2016).[29,36] Chris had seven bouts after his surgery.[38] In his most recent bout on 04/24/21, Chris suffered a tibia-fibular fracture for which he underwent surgery, but plans to return to fighting in 07/2023.[39] Nathan underwent a CDR for disk herniation on 09/14/2015 and successfully returned to play the following season.[37] His stats after surgery seem comparable with those before the surgery although he did not play because many games the last 2 seasons before eventually retiring because of injuries unrelated to the spine.[40,41] Performance statistics of these two athletes are presented in **Supplemental Table 3d-e**, http://links.lww.com/NEU/D906.

DISCUSSION

Given the controversial nature of TDRs in elite athletes, a systematic review of the scientific literature was performed. Seven studies reported CDRs in 53 athletes, of which seven were professional and one played contact sports. Three studies reported LDRs in 51 athletes, of which 17 were professional and 8 played contact sports. All professional athletes successfully returned to play without complications. A search of the lay press revealed six professional athletes who underwent CDR, five of whom played contact sports. There has been no report of a TDR in professional American football to date. The current review represents the first attempt to summarize the use of TDR in athletes.

All seven professional athletes who underwent a CDR from the scientific literature returned to play successfully; however, only one played a contact sport (kickboxing). While our review of the scientific literature suggests that CDR may be safe and effective for limited-contact/noncontact sports athletes, further investigation is needed for those participating in contact sports. Previous studies have shown that athletes with one-level and two-level anterior cervical diskectomy and fusion (ACDF) may return to contact sports safely in most cases.[14,42-45] In a recent modified Delphi consensus study with the Cervical Spine Research Society, Schroeder et al[45] presented an 84% consensus among 65 that asymptomatic athletes without increased magnetic

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resonance imaging T2 signal changes after 1-level or 2-level ACDF may RTP, though not after 3-level ACDF. In addition, 99% of physicians agreed that asymptomatic athletes with a solid fusion after a compression, burst, or facet fracture without instability or T2-signal change were allowed to RTP. However, primary studies and consensus guidelines/recommendations for RTP in the setting of CDR in contact/collision sports athletes remain highly limited.[11,14,15,20,24,43] Unfortunately, the scientific literature reports only one contact of sport athlete who returned to play after CDR, a kickboxer who completed the final year of her career.[13] Perhaps the best evidence we have in elite athletes comes from the three NHL players, one mixed martial arts (MMA) fighter, and one Australian-rules football player and is that they each successfully returned to play.

While LDR is much less common in the United States than CDR, our review found several international studies presenting RTP data in professional athletes after LDR. Three studies discussed LDR in 51 athletes; 17 were professional, eight of whom played contact sports (2 boxing, 2 alpine skiing, 2 soccer, 1 judo, 1 rugby) and all returned to play successfully. The current literature suggests that LDR may be a safe and effective surgery for professional athletes participating in contact sports. The differences in anatomical considerations between CDR and LDR, such that a complication regarding a CDR may result in paralysis and quadriplegia,[43] make LDR a less controversial treatment. A nerve root complication, leading to pain and/or weakness,[46] is certainly a problem, but less concerning than a cervical spinal cord injury. Given the variance in risk profiles, outcomes and/or recommendations of LDR need to be further explored, especially in the United States. Moreover, efficacy and safety of the anterior- vs posterior-based procedure are of particular interest.

The lay press search revealed five professional contact sport athletes, all of whom successfully returned to play to their previous level of competition. The athletes included three hockey players in the NHL,[16,28,35] one mixed martial arts fighter in the UFC,[29] and one Australian-rules football player in the Australian Football League.[31] Across all five athletes, there were no documented complications. The Australian-rules football player retired after 4 successful years of play after surgery, and the three NHL hockey players and the MMA fighter continued to play at the professional level in their corresponding organizations. From the publicly available data, it seems that CDR is safe in professional contact sport athletes; however, one must question the validity of publicly reported data. Public data are not subjected to the same rigorous processes that peer-reviewed publications undergo and thus have limitations in providing clinical guidance.[14] Previous reports have found that publicly reported injuries can be inaccurate.[47] Further efforts are needed to fill this major gap in the current literature, with the goal of providing consensus guidelines following TDR in elite athletes.[11,14,15,20,24,43] It is our hope that this review serves as the groundwork for future studies on the topic of TDR in elite contact sport athletes.

Limitations

The current study has several limitations. First, some information including key demographics, surgery, and outcomes was missing from the included studies. Regarding outcomes, besides RTP data, there might have been heterogeneity in definitions of symptom resolution, complications, and patient satisfaction. In addition, the length of follow-up of the athletes was limited. The mean follow-up ranged significantly for CDR (0.1-7.0 years) and was only a few years for LDR (2.0-3.2 years). It is also important to note that in some studies that grouped TDRs, fusions, and diskectomies together, the values of certain quantitative variables (eg, mean follow-up) represented those of the entire sample. However, these limitations were inevitable because the role of a systematic review is to summarize information that is present in the literature. Our review, nonetheless, shows that TDRs are being performed in athletes who are successfully returning to play. Second, we included a review of publicly available data on elite athletes and TDR surgery. It is possible that publicly reported information was false because we did not have access to their medical records. Moreover, follow-up data from the lay press on professional athletes were limited for two reasons: (1) the CDRs were performed recently and many of these athletes (4/5) are still competing professionally, and (2) obtaining meaningful data from the lay press is challenging. Finally, the categorization of contact vs limited-contact and noncontact sports was adopted from a previous study,[17] and we recognize that other classifications may exist.

CONCLUSION

We identified 10 studies discussing TDR in elite athletes. Of the 53 athletes who underwent CDR, one played professional contact sports. Of the 51 athletes who underwent LDR, eight played professional contact sports. The lay press revealed five professional contact sport athletes who underwent a CDR, all with successful RTP. From the little data that exist, TDR seems to be safe in elite athletes; however, the small number of patients highlights the paucity of data in this area, particularly in contact sports.

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Supplementary Table 1. Search terms.
Supplementary Table 2. Levels of Evidence for Primary Research Question as Adopted by the North American Spine Society (NASS).
Supplementary Table 3. Performance statistics of professional contact sport athletes before and after cervical disc replacement.

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